Experiment 6:

CODE:

```
 GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main(int argc, char *argv[])
{
    int mb;
    if (argc != 2)
    {
        printf("Usage: %s <memory_in_MB>\n", argv[0]);
        return 1;
    }

    mb = atoi(argv[1]);
    char *memory = malloc(mb * 1024 * 1024);
    if (memory == NULL)
    {
        printf("Memory allocation failed\n");
        return 1;
    }

    for (int i = 0; i < mb * 1024 * 1024; i += 4096)
    {
        memory[i] = 1;
    }

    printf("%d MB memory allocated.\n", mb);
    printf("Program is running... Press Ctrl+c to exit.\n");

    while(1)
    {
        sleep(1);
    }
    return 0;
}
```

OUTPUT:

```
tofiq@Tofiq:~$ ./exp6 200
200 MB memory allocated.
Program is running... Press Ctrl+c to exit.
^C
```

```
tofiq@Tofiq:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           4.3Gi       1.4Gi       1.0Gi        50Mi       2.1Gi       2.9Gi
Swap:             0B          0B          0B
tofiq@Tofiq:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           4.3Gi       1.2Gi       1.2Gi        50Mi       2.1Gi       3.0Gi
Swap:             0B          0B          0B
```